Per-site ΔRMSE from adding disturbance — M3 (C+T) vs M4 (C+T+D)

Below 0 = D reduces the site's RMSE | bars coloured by relative disturbance class (site maximum) | repeated CV (3×80 %) | tree cover ≥30 % (104 sites, 381 site-years) · deadtrees v4.1

ΔRMSE = RMSE(with D) − RMSE(without D)

GBoost (Optuna) · 12r | GBoost (Optuna) · 24r | ghtGBM (Optuna) · 12 | ghtGBM (Optuna) · 24 | toGluon (ensemble) · 1 | toGluon (ensemble) · 2

GPPsat (µmol m⁻² s⁻¹)

62% improved | 60% improved | 64% improved | 62% improved | 58% improved | 60% improved

NEPmax (µmol m⁻² s⁻¹)

74% improved | 73% improved | 73% improved | 70% improved | 62% improved | 64% improved

ETmax (mm d⁻¹)

57% improved | 56% improved | 59% improved | 53% improved | 53% improved | 60% improved

WUE (g C mm⁻¹)

54% improved | 67% improved | 57% improved | 55% improved | 61% improved | 55% improved

Sites (sorted by ΔRMSE within each panel)

Relative disturbance
site maximum, mean of 100-500 m buffers
(breaks: mean ± 0.5 SD)

- Low (< 12.4 %, 40 sites)
- Mid (12.4–25.0 %, 43 sites)
- High (> 25.0 %, 21 sites)